

PHYS 143 – Mechanics and Thermal Physics

Work and Kinetic Energy

How do forces change an object's motion as it moves through space?

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Reading: Young and Freedman, *University Physics* (15th ed.), Ch. 6, §6.1–6.4

By the end of this week, you should be able to:

1. Calculate the work done by a constant force, and the total work done on an object.
2. Interpret the sign of work and the meaning of kinetic energy.
3. Apply the work–energy theorem to find speeds, distances and forces.
4. Calculate the work done by varying forces, including springs, and relate work to power.

The detailed outcomes are in the checklist at the end of the lecture.

Three contacts:

1 IDENTIFY

When is work done, and how much?

2 QUANTIFY

How does work change motion?

3 EXTEND

How do varying forces and power fit in?

Not marked. Answer individually, then we review.

1. Find $\vec{A} \cdot \vec{B}$ for $A = 3.0$ at 0° and $B = 4.0$ at 60° .
2. Find the area under a straight line from $(0, 0)$ to $(4, 6)$, and the area of a 3 by 5 rectangle.
3. A car accelerates uniformly from 10 to 20 m/s over 50 m. Find its acceleration.
4. A 10 kg box is pulled by a 50 N horizontal force against 20 N of friction. Find its acceleration.
5. A 5.0 kg block slides on a rough floor with $\mu_k = 0.30$. Find the friction force.
6. Convert 3000 rev/min to rad/s.

1. $\vec{A} \cdot \vec{B} = AB \cos 60^\circ = 6.0.$

2. $\frac{1}{2}(4)(6) = 12; 3 \times 5 = 15.$

3. $a = (20^2 - 10^2)/(2 \times 50) = 3.0 \text{ m/s}^2.$

4. $a = (50 - 20)/10 = 3.0 \text{ m/s}^2.$

5. $f = \mu_k mg = 0.30(5.0)(9.8) = 14.7 \text{ N}.$

6. $3000 \times 2\pi/60 = 314 \text{ rad/s}.$

Week 1 (dot product), Week 2 (areas under graphs, constant acceleration) and Week 3 (free-body diagrams, friction) are all reused this week. A short calculus refresher is in the supplementary notes.

Contact 1: Identify

When is work done, and how much?

From forces to forces acting over a distance

Week 3

$\sum \vec{F} = m\vec{a}$ at one instant.

What accelerates an object?

Week 4

a force acting *through a displacement*.

What does it do to the object's speed?

Opening question

In which of these is work done **on the load** by the person?

- (a) A weightlifter holds a barbell motionless overhead.
- (b) A porter walks along level ground carrying a bag at constant speed.
- (c) A person lifts a box from the floor onto a shelf.

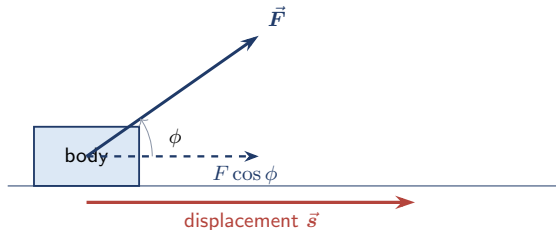
Predict first

Decide before the next slides. Which situations involve a displacement of the load?
In which direction is the person's force, compared with the displacement?

Rule: work done by a constant force

$$W = \vec{F} \cdot \vec{s} = F s \cos \phi$$

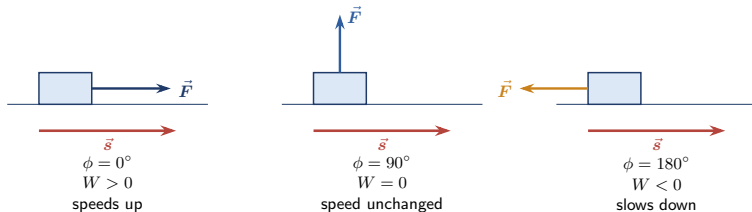
ϕ is the angle between the force $\vec{F}$ and the displacement $\vec{s}$. Work is a **scalar**.



Physical meaning

Only the component of the force **along the displacement**, $F \cos \phi$, does work. This is the dot product from Week 1: “how much of $\vec{F}$ points along $\vec{s}$?”.

The sign of work



Rule: sign of work

$$0 \leq \phi < 90^\circ: W > 0 \quad \phi = 90^\circ: W = 0 \quad 90^\circ < \phi \leq 180^\circ: W < 0$$

Unit: $1 \text{ J} = 1 \text{ N m} = 1 \text{ kg m}^2/\text{s}^2$. Positive work is done *on* the object by the force; negative work means the force opposes the motion.

TOOLKIT 1 – Work by a constant force

1. **IDENTIFY** the force and the displacement of the *object it acts on*. Sketch both.
2. **ANGLE:** find ϕ between $\vec{F}$ and $\vec{s}$ (tail to tail).
3. **PREDICT** the sign of W from ϕ .
4. **CALCULATE:** $W = Fs \cos \phi$, or $W = F_x s_x + F_y s_y$ in components (Week 1).
5. **UNITS:** newton $\times$ metre = joule.

RULE: only the component of the force along the displacement does work.

WATCH OUT: a force perpendicular to the displacement does no work; no displacement means no work by that force.

CHECK: does the sign of the answer agree with your prediction from the sketch?

Question

A person pulls a trolley 15.0 m along level ground with a rope whose tension is 120 N, at 35° above the horizontal. How much work does the rope's force do on the trolley?

Predict first

Positive, zero or negative? Roughly: less or more than $120 \times 15 = 1800 \text{ J}$?

Question

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Predict first

Positive, zero or negative? Roughly: less or more than $120 \times 15 = 1800$ J?

Identify, set up: $+x$ along the displacement; $\phi = 35^\circ$.

Execute (angle form): $W = Fs \cos \phi = (120)(15.0) \cos 35^\circ = 1.47 \times 10^3$ J.

Execute (component form): $F_x = 120 \cos 35^\circ = 98.3$ N, $F_y = 69$ N, $\vec{s} = 15.0 \hat{i}$:

$W = F_x s_x + F_y s_y = (98.3)(15.0) + 0 = 1.47 \times 10^3$ J.

Evaluate: positive, and less than 1800 J because only part of the force is along $\vec{s}$. The vertical component does no work. ✓

Concept check: which forces do no work?

Does the named force do work? (Give the sign, or zero.)

1. The weightlifter's force on the barbell while holding it overhead.
2. The porter's upward force on the bag while walking at constant speed on level ground.
3. The normal force of a table on a block sliding across it.
4. The tension in the string of a ball moving in a circle.

Predict first

For each: what are $\vec{F}$ and $\vec{s}$, and what is ϕ ?

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Predict first

For each: what are $\vec{F}$ and $\vec{s}$, and what is ϕ ?

All four do zero work. (1) $s = 0$. (2), (3), (4): the force is perpendicular to the displacement, $\phi = 90^\circ$.

Common error: “it takes effort, so work is done”. In physics, work needs a displacement along the force.

Is a hard “workout” work?

Lifting and lowering

You lift a 20 kg box 1.0 m at constant speed, then lower it 1.0 m at constant speed. How much work did your upward force do in total?

Lifting: $\phi = 0^\circ$, $W = +mgh = +196 \text{ J}$.

Lowering: $\phi = 180^\circ$, $W = -196 \text{ J}$.

Total: 0.

Why does it still feel like work?

Your muscles use chemical energy even when the net work on the box is zero: they are tensing and relaxing internally. Physics “work” counts only the energy transferred to the *object* by a force along its displacement.

Total work on an object

Rule: total (net) work

When several forces act on an object, the total work can be found two ways:

Method I: add the work done by each force: $W_{\text{tot}} = W_1 + W_2 + W_3 + \dots$ (an *algebraic* sum, with signs).

Method II: find the net force $\sum \vec{F}$ and use it: $W_{\text{tot}} = (\sum \vec{F}) \cdot \vec{s}$.

Predict first

Which forces on a sliding box do work? Which do not? (Week 3 free-body diagram.)

TOOLKIT 2 – Net work from the free-body diagram

- 1. FBD:** draw every force on the object (Week 3 Toolkit 1).
- 2. DISPLACEMENT:** draw $\vec{s}$ on the same sketch.
- 3. LIST** each force, find ϕ for each, and assign a sign: $W > 0$, $W = 0$ or $W < 0$.
- 4. CALCULATE** each work (Toolkit 1).
- 5. METHOD I:** add them with signs.
- 6. METHOD II (check):** find the net force along $\vec{s}$ and multiply by s .

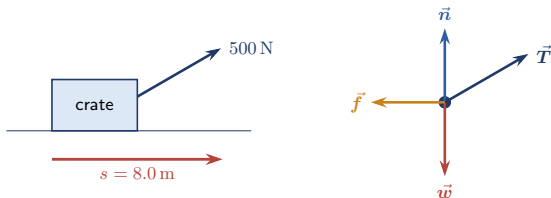
WATCH OUT: normal force and weight on a horizontal floor do no work; friction and tension may.

CHECK: do Method I and Method II agree?

Lecturer example: a tractor pulls a crate

Question

A tractor pulls a 50.0 kg crate 8.0 m across level ground with a cable force of 500 N at 30° above the horizontal. The coefficient of kinetic friction is $\mu_k = 0.20$. Find the work done by each force and the total work.



Predict first

Which forces do positive work, negative work, and no work? Is the total positive?

The tractor and crate: solution

Set up: $+x$ along $\vec{s}$. Vertical: $n = mg - T \sin 30^\circ = 490 - 250 = 240 \text{ N}$; so $f_k = \mu_k n = 0.20(240) = 48 \text{ N}$ (Week 3).

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Method I:

$$W_T = Ts \cos 30^\circ = (500)(8.0)(0.866) = +3460 \text{ J} \quad W_f = -f_k s = -(48)(8.0) = -384 \text{ J}$$
$$W_w = W_n = 0 \text{ (perpendicular to } \vec{s}) \quad W_{\text{tot}} = 3460 - 384 = \mathbf{3080 \text{ J}}$$

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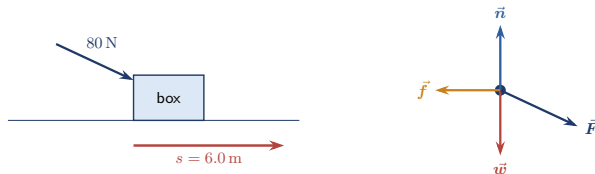
Method II (check): net force along x : $500 \cos 30^\circ - 48 = 385 \text{ N}$;
 $W_{\text{tot}} = (385)(8.0) = 3080 \text{ J}$. ✓

Evaluate: the signs match the prediction; both methods agree.

Your turn: pushing a box downward at an angle

Your turn

A 20.0 kg box is pushed 6.0 m along a floor by a force of 80 N directed 25° **below** the horizontal. The coefficient of kinetic friction is 0.30. Find the work done by each force and the total work.



Predict first

Does the downward component change the friction? Is the total work large or small compared with the work of the push alone?

Set up: $n = mg + F \sin 25^\circ = 196 + 80(0.423) = 229.8 \text{ N}$, so $f_k = 0.30(229.8) = 68.9 \text{ N}$.

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Execute:

$$W_F = Fs \cos 25^\circ = (80)(6.0)(0.906) = +435 \text{ J} \quad W_f = -(68.9)(6.0) = -414 \text{ J}$$

$$W_w = W_n = 0$$

$$W_{\text{tot}} = 435 - 414 = +\mathbf{21 \text{ J}}$$

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Evaluate: Method II: $(80 \cos 25^\circ - 68.9)(6.0) = (72.5 - 68.9)(6.0) = 21 \text{ J}$. ✓

The push is mostly cancelled by friction, so the total is small. Pushing downward *increases* n , and hence the friction.

Contact 1: what we can now do

Force and displacement



Work $W = \vec{F} \cdot \vec{s} = F s \cos \phi$



Total work (Methods I and II)

You should be able to

- calculate work with $F s \cos \phi$;
- decide the sign of work from the angle;
- list which forces do work on a free-body diagram;
- find the total work two ways.

✓ Before Contact 2

If the total work on an object is positive, what happens to its speed? Read Young and Freedman, §6.2.

Contact 2: Quantify

How does work change motion?

Poll

A car travelling at 25 m/s stops with a constant braking force over a distance d . If it were travelling at 50 m/s and the same braking force were applied, what would the stopping distance be?

Predict first

Twice d ? Four times d ? Eight times? Commit to an answer; we will derive it.

Idea

When a net force acts on a moving object over a displacement:

$W_{\text{tot}} > 0$: it **speeds up**

$W_{\text{tot}} < 0$: it **slows down**

$W_{\text{tot}} = 0$: its speed **stays the same**

Predict first

A constant net force acts along the motion. Which quantity connects the force, the distance and the speed? (Hint: Week 2.)

We now show why, using Newton's second law and the constant-acceleration equation from Week 2.

Constant net force along a straight line. Take $+x$ along the motion.

1. Newton's second law: $\sum F_x = ma_x$.
2. Week 2: $v^2 = v_0^2 + 2a_x(x - x_0)$, so $a_x = \frac{v^2 - v_0^2}{2(x - x_0)}$.
3. Substitute: $\sum F_x(x - x_0) = m \frac{v^2 - v_0^2}{2} = \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2$.

The left side is the total work W_{tot} . Define the **kinetic energy** $K = \frac{1}{2}mv^2$:

$$W_{\text{tot}} = K_2 - K_1 = \Delta K$$

Rule: the work–energy theorem

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$$W_{\text{tot}} = K_2 - K_1 = \Delta K \quad \text{with} \quad K = \frac{1}{2}mv^2$$

The total (net) work done on a particle equals the change in its kinetic energy.

- K is a **scalar** and never negative. Unit: joule.
- $K \propto v^2$: doubling the speed quadruples K .
- It is the *net* work, not the work of one force.

Model assumptions

The object is a particle (or moves without rotating); the theorem uses all forces, including friction. It holds for varying forces too (shown in the supplement).

TOOLKIT 3 – Applying the work–energy theorem

1. **STATES:** choose an initial point (1) and a final point (2); mark both on the sketch with the speeds.
2. **FBD:** list every force on the object between 1 and 2 (Week 3), and the displacement.
3. **WORK:** find the work of each force with its sign (Toolkits 1 and 2); add for W_{tot} .
4. **WRITE** $W_{\text{tot}} = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$ and solve for the unknown (a speed, a distance or a force).

WATCH OUT: the work of the net force, not of one force; forces that do no work still appear in the free-body diagram.

CHECK: does the sign of ΔK agree with whether the object speeds up or slows down?

Question

A 10.0 kg crate moving at 2.0 m/s on a level floor is pushed 8.0 m by a constant horizontal force of 50 N. Kinetic friction exerts 20 N. What is its speed after the 8.0 m?

Predict first

Speeds up or slows down? Roughly how much work does the net force do?

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Predict first

Speeds up or slows down? Roughly how much work does the net force do?

Set up: $+x$ along the motion. **Work:** $W_{\text{push}} = (50)(8.0) = +400 \text{ J}$,
 $W_f = -(20)(8.0) = -160 \text{ J}$, $W_w = W_n = 0$, so $W_{\text{tot}} = 240 \text{ J}$.

Execute: $K_1 = \frac{1}{2}(10.0)(2.0)^2 = 20 \text{ J}$, so $K_2 = 20 + 240 = 260 \text{ J}$ and
 $v_2 = \sqrt{2(260)/10.0} = 7.2 \text{ m/s}$.

Evaluate: $W_{\text{tot}} > 0$ and $v_2 > v_1$, as predicted. ✓

The same problem by Newton's second law:

$$a = \frac{50 - 20}{10.0} = 3.0 \text{ m/s}^2$$

$$v^2 = v_0^2 + 2a\Delta x = 4.0 + 2(3.0)(8.0) = 52$$

$$v = 7.2 \text{ m/s} \quad \checkmark$$

Both methods agree. Which was shorter? Which gives the *time*?



TOOLKIT 4 – Choosing the method

ENERGY METHOD: the question is about *speed, distance or force* between two positions, and the time is not needed.

NEWTON'S SECOND LAW: the question is about *acceleration or time*, or the motion is along a path where only the net force direction matters.

WATCH OUT: energy gives no direction of motion and no time.

CHECK: solve a simple case both ways.

Question

A 1200 kg car at 25.0 m/s stops with a constant braking force of 7200 N. Find the stopping distance. What if it started at 50.0 m/s?

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$$W_{\text{tot}} = -Fd = 0 - \frac{1}{2}mv_1^2 \Rightarrow d = \frac{mv_1^2}{2F}.$$

$$25.0 \text{ m/s: } d = \frac{1200(625)}{2(7200)} = 52 \text{ m} \quad (\text{the same answer as the Week 2 braking example}).$$

$$50.0 \text{ m/s: } d = \frac{1200(2500)}{2(7200)} = 208 \text{ m, } \mathbf{\text{four times}} \text{ the distance.}$$

Physical meaning

$d \propto v^2$ because $K \propto v^2$: the force must remove four times as much kinetic energy.

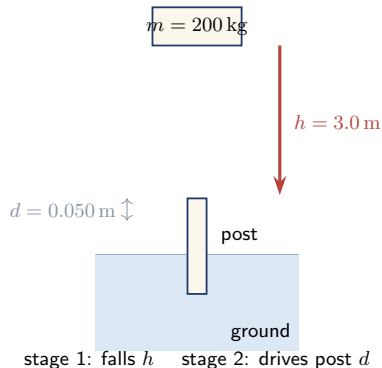
Rule: work by gravity (near the Earth's surface)

An object of weight mg that moves **down** through a height h has work done on it by gravity of $W_w = +mgh$. Moving **up** through h : $W_w = -mgh$.

This is just $W = \vec{F} \cdot \vec{s}$ with $\vec{F} = m\vec{g}$ downward. Only the vertical change in position matters, not the path. (We will use this idea to define potential energy in the next lecture.)

Question

A 200 kg hammerhead is released from rest 3.0 m above a post and falls freely onto it, then drives the post 0.050 m into the ground. Find (a) the hammer's speed at impact, and (b) the average force on the post while it is driven in.



Predict first

Is the average force on the post larger or smaller than the hammer's weight, 1960 N? By roughly how much?

(a) Stage 1: states: released at rest ($K_1 = 0$), just before impact. Only gravity does work:

$$W_w = +mgh = (200)(9.8)(3.0) = 5880 \text{ J.}$$

$$\frac{1}{2}mv^2 = 5880 \Rightarrow v = \sqrt{2(5880)/200} = 7.7 \text{ m/s.}$$

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(b) Stage 2: from impact ($K = 5880 \text{ J}$) to rest. Forces: weight (down, $+mgd$) and the post's resistive force F (up, $-Fd$).

$$W_{\text{tot}} = mgd - Fd = 0 - 5880 \Rightarrow F = \frac{5880}{0.050} + mg = 1.18 \times 10^5 + 1.96 \times 10^3 = 1.2 \times 10^5 \text{ N.}$$

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Evaluate: about 60 times the weight, as predicted. Check over the whole motion:

$$mg(h + d) = Fd \Rightarrow F = \frac{1960(3.05)}{0.050} = 1.2 \times 10^5 \text{ N. } \checkmark$$

The physical meaning of kinetic energy

From part (a) of the pile driver

K is the total work done on an object to bring it **from rest to its present speed**.

From part (b)

K is also the total work the object can do in being **brought to rest**.

Physical meaning

Kinetic energy is the energy of motion. The work–energy theorem tells us how the energy of motion changes when forces do work.

Predict first

Which has more kinetic energy: a 1000 kg car at 20 m/s, or a 2000 kg truck at 10 m/s?

Comparing kinetic energies

Car: $K = \frac{1}{2}(1000)(20)^2 = 2.0 \times 10^5 \text{ J}$ **Truck:** $K = \frac{1}{2}(2000)(10)^2 = 1.0 \times 10^5 \text{ J}$.

The car has **twice** the kinetic energy of the truck, although the truck has twice the mass.

Two iceboats finish with equal K

One has twice the mass of the other. What is the ratio of their speeds?

Comparing kinetic energies

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The car has **twice** the kinetic energy of the truck, although the truck has twice the mass.

Two iceboats finish with equal K

One has twice the mass of the other. What is the ratio of their speeds?

$$\frac{1}{2}m_1v_1^2 = \frac{1}{2}m_2v_2^2 \text{ with } m_1 = 2m_2:$$

$v_2/v_1 = \sqrt{2}$. The lighter boat moves $\sqrt{2}$ times faster.

Common error

Doubling the speed does not double K ; it multiplies it by 4. Doubling the mass doubles K .

Concept check: can friction do positive work?

Question

A 20 kg box rests on a lorry bed. The lorry accelerates at 2.0 m/s^2 over 50 m and the box does not slide. Find the work done on the box by static friction, and compare it with the box's change in kinetic energy.

Predict first

Which way does friction point on the box? Is its work positive or negative?

Concept check: can friction do positive work?

Question

A 20 kg box rests on a lorry bed. The lorry accelerates at 2.0 m/s^2 over 50 m and the box does not slide. Find the work done on the box by static friction, and compare it with the box's change in kinetic energy.

Predict first

Which way does friction point on the box? Is its work positive or negative?

$f_s = ma = (20)(2.0) = 40 \text{ N}$ in the direction of motion, so $W = +(40)(50) = +2000 \text{ J}$. The box starts from rest and ends at $v^2 = 2(2.0)(50) = 200$, so $\Delta K = \frac{1}{2}(20)(200) = 2000 \text{ J}$. ✓

Friction can do positive work, negative work or none; it depends on the angle between the force and the displacement.

Your turn: a block sliding to rest

Your turn

A 2.0 kg block slides on a rough floor with initial speed 6.0 m/s. The coefficient of kinetic friction is 0.25. How far does it slide before stopping?

Predict first

Sketch. Which force does the work that changes K ? Is the distance more or less than 5 m?

Your turn

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Predict first

Sketch. Which force does the work that changes K ? Is the distance more or less than 5 m?

$$f_k = \mu_k mg = 0.25(2.0)(9.8) = 4.9 \text{ N}; \quad W_{\text{tot}} = -f_k d = 0 - \frac{1}{2}mv_1^2 = -36 \text{ J}$$
$$d = 36/4.9 = 7.3 \text{ m}.$$

Evaluate: more than 5 m, as predicted; the mass cancels: $d = v_1^2/(2\mu_k g)$. ✓

Total work

$$\Downarrow W_{\text{tot}} = \Delta K$$

$$\text{Kinetic energy } K = \frac{1}{2}mv^2$$

$\Downarrow$

Speed, distance or force

You should be able to

- derive and state $W_{\text{tot}} = \Delta K$;
- explain why $K \propto v^2$;
- apply the theorem to speeds, distances and forces;
- choose between the energy method and Newton's second law.

✓ Before Contact 3

Read Young and Freedman, §6.3–6.4. What happens when the force is *not* constant, for example a stretched spring?

Contact 3: Extend

How do varying forces and power fit in?

Opening question

When you drive a car, the force on it keeps changing as you use the accelerator and the brake. We cannot use $W = Fs \cos \phi$ with one value of F . **How do we find the work?**

Predict first

Recall Week 2: how did we find the displacement when the velocity was not constant?

Opening question

When you drive a car, the force on it keeps changing as you use the accelerator and the brake. We cannot use $W = Fs \cos \phi$ with one value of F . **How do we find the work?**

Predict first

Recall Week 2: how did we find the displacement when the velocity was not constant?

The same idea: split the path into tiny steps Δx over which F_x is almost constant, add up $F_x \Delta x$, and take the limit.

Rule: work by a varying force (straight line)

$$W = \int_{x_1}^{x_2} F_x \, dx = \text{area under the } F_x\text{-}x \text{ graph}$$

For a constant F_x , $W = F_x(x_2 - x_1)$, the rectangle.

- Area above the axis: $W > 0$. Area below the axis (force opposing the motion): $W < 0$.
- The same idea as “area under the v - t graph = displacement” in Week 2.

TOOLKIT 5 – Work from graphs and integrals

1. **SKETCH** F_x against x for the motion.
 2. **SPLIT** the region into rectangles and triangles.
 3. **AREAS**: each area is a work in joules; areas below the axis are negative.
 4. **INTEGRATE** if $F_x(x)$ is a formula: $W = \int F_x dx$, with the limits as the start and end positions.
- WATCH OUT:** limits are positions x_1 and x_2 , not times; check the units ($\text{N} \times \text{m}$).
- CHECK:** does the integral agree with the area of a sketch?

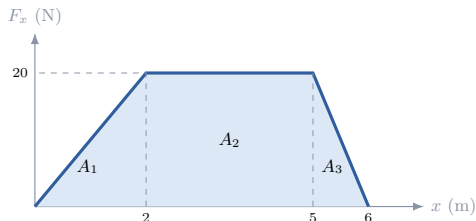
Example: work from an F_x - x graph

Question

A force on a particle varies with position as shown. Find the work it does as the particle moves from $x = 0$ to $x = 6.0$ m.

Predict first

Rough size: the maximum force is 20 N over 6 m; is the work more or less than 120 J?



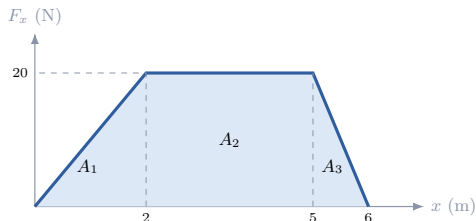
Example: work from an F_x - x graph

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Predict first

Rough size: the maximum force is 20 N over 6 m; is the work more or less than 120 J?



$$A_1 = \frac{1}{2}(2.0)(20) = 20 \text{ J}, \quad A_2 = (3.0)(20) = 60 \text{ J}, \quad A_3 = \frac{1}{2}(1.0)(20) = 10 \text{ J}. \quad W = \mathbf{90 \text{ J}}$$

(less than 120 J, as predicted).

Your turn

The force on a particle is $F_x = 3.0x^2$ (SI units, force in newtons, x in metres). Find the work done as it moves from $x = 0$ to $x = 2.0$ m.

Predict first

The force rises from 0 to 12 N. Is the work more or less than $12 \times 2.0 = 24$ J?

Your turn

The force on a particle is $F_x = 3.0x^2$ (SI units, force in newtons, x in metres). Find the work done as it moves from $x = 0$ to $x = 2.0$ m.

Predict first

The force rises from 0 to 12 N. Is the work more or less than $12 \times 2.0 = 24$ J?

$$W = \int_0^{2.0} 3.0x^2 \, dx = \left[x^3 \right]_0^{2.0} = 8.0 \, \text{J}.$$

Evaluate: less than 24 J, as predicted, because the force is small over most of the distance. ✓

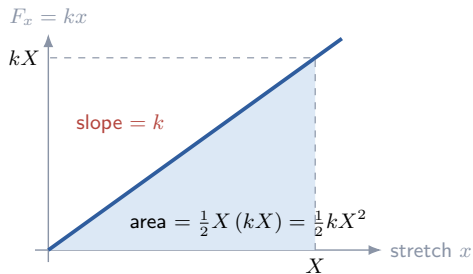
The spring force

Rule: Hooke's law

The force needed to hold a spring stretched (or compressed) by x from its natural length is

$$F_x = kx$$

k is the spring constant, in N/m. The force *by* the spring on the object is $-kx$ (it opposes the displacement).



Model assumptions

Hooke's law holds for small stretches; beyond the elastic limit the force is no longer proportional to x .

Rule: work on a spring

Stretching a spring from $x = 0$ to $x = X$ requires the work (the area of the triangle):

$$W = \int_0^X kX' dX' = \frac{1}{2}kX^2$$

More generally, from x_1 to x_2 : $W = \frac{1}{2}kx_2^2 - \frac{1}{2}kx_1^2$.

Careful with signs

The work done *on* the spring by you is positive. The work done *by* the spring on the object that stretches it is the **negative** of this: $W_{\text{spring}} = -\frac{1}{2}kX^2$ when stretching it from natural length.

TOOLKIT 6 – Springs

1. **NATURAL LENGTH:** measure x from the unstretched (or uncompressed) position.
2. **DIRECTION:** the spring's force on the object points back toward $x = 0$: $F = -kx$.
3. **WORK:** $W_{\text{by spring}} = \frac{1}{2}kx_1^2 - \frac{1}{2}kx_2^2$ between x_1 and x_2 ; the work you do on the spring is the negative.
4. **COMBINE** with other forces using the work–energy theorem (Toolkit 3).

WATCH OUT: use the stretch from natural length, not the spring's total length; signs of the work by and on the spring.

CHECK: is k in N/m and x in m?

Question

A 60.0 kg person stands on a bathroom scale. Its spring compresses by 0.80 cm. Find the spring constant and the work done on the spring.

Predict first

Is k of order 10^2 , 10^4 or 10^6 N/m? Is the work of order 1 J or 100 J?

Question

A 60.0 kg person stands on a bathroom scale. Its spring compresses by 0.80 cm. Find the spring constant and the work done on the spring.

Predict first

Is k of order 10^2 , 10^4 or 10^6 N/m? Is the work of order 1 J or 100 J?

At rest, the spring force balances the weight: $F = mg = 588$ N at $x = 0.0080$ m.

$k = F/x = 588/0.0080 = 7.4 \times 10^4$ N/m. Work on the spring:

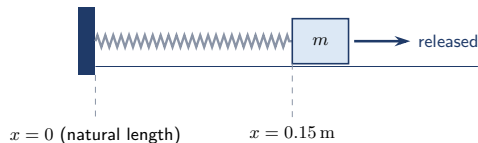
$$\frac{1}{2}kx^2 = \frac{1}{2}(588)(0.0080) = 2.4 \text{ J.}$$

Evaluate: of order 10^4 N/m and about 2 J, as predicted. ✓

Lecturer example: a glider on a spring

Question

A 0.200 kg glider on a horizontal air track is attached to a spring ($k = 40.0 \text{ N/m}$) and pulled 0.15 m from natural length, then released from rest. Find its speed when it passes $x = 0$ (a) with no friction, and (b) with a constant friction force of 0.50 N.



Predict first

Is the speed in (b) larger or smaller than in (a)?

A glider on a spring: solution

Set up: states: released ($x_1 = 0.15 \text{ m}$, $v_1 = 0$) and at $x = 0$.

$$W_{\text{spring}} = \frac{1}{2}kx_1^2 - 0 = \frac{1}{2}(40.0)(0.15)^2 = 0.45 \text{ J}.$$

A glider on a spring: solution

Set up: states: released ($x_1 = 0.15$ m, $v_1 = 0$) and at $x = 0$.

$$W_{\text{spring}} = \frac{1}{2}kx_1^2 - 0 = \frac{1}{2}(40.0)(0.15)^2 = 0.45 \text{ J.}$$

(a) No friction: $W_{\text{tot}} = 0.45 = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2(0.45)/0.200} = 2.1 \text{ m/s.}$

A glider on a spring: solution

Set up: states: released ($x_1 = 0.15$ m, $v_1 = 0$) and at $x = 0$.

$$W_{\text{spring}} = \frac{1}{2}kx_1^2 - 0 = \frac{1}{2}(40.0)(0.15)^2 = 0.45 \text{ J.}$$

(a) No friction: $W_{\text{tot}} = 0.45 = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2(0.45)/0.200} = 2.1 \text{ m/s.}$

(b) Friction: $W_f = -(0.50)(0.15) = -0.075 \text{ J, so } W_{\text{tot}} = 0.45 - 0.075 = 0.375 \text{ J:}$
 $v = \sqrt{2(0.375)/0.200} = 1.9 \text{ m/s.}$

A glider on a spring: solution

Set up: states: released ($x_1 = 0.15 \text{ m}$, $v_1 = 0$) and at $x = 0$.

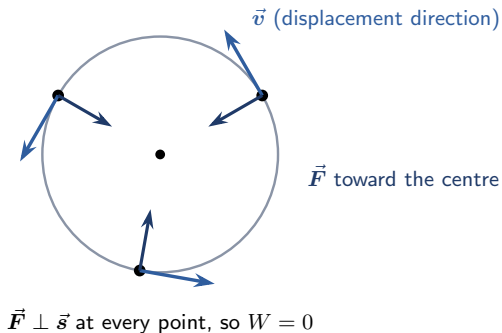
$$W_{\text{spring}} = \frac{1}{2}kx_1^2 - 0 = \frac{1}{2}(40.0)(0.15)^2 = 0.45 \text{ J}.$$

(a) No friction: $W_{\text{tot}} = 0.45 = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2(0.45)/0.200} = 2.1 \text{ m/s}$.

(b) Friction: $W_f = -(0.50)(0.15) = -0.075 \text{ J}$, so $W_{\text{tot}} = 0.45 - 0.075 = 0.375 \text{ J}$:
 $v = \sqrt{2(0.375)/0.200} = 1.9 \text{ m/s}$.

Evaluate: (b) is smaller, as predicted. The spring force is not constant, but the work–energy theorem still works because we only need the *work* between the two positions. ✓

Motion on a curved path



Rule: perpendicular force

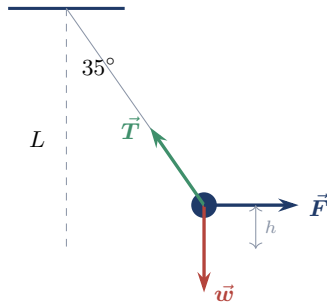
A force that is always perpendicular to the displacement does **no work**.

Examples: the tension in a pendulum string, the normal force on a sliding block, the centripetal force in uniform circular motion. So $W_{\text{tot}} = 0$ and K , hence the speed, is unchanged (consistent with Weeks 2 and 3).

Lecturer example: pushing a child on a swing

Question

A 25.0 kg child sits on a swing of length 3.30 m. You pull the child slowly (constant speed, in equilibrium at every point) with a horizontal force until the chain makes 35° with the vertical. How much work do you do?



Predict first

Which of F , T and w do work? What is W_{tot} ?

A child on a swing: solution

Set up: the child moves slowly, so K is almost constant: $W_{\text{tot}} = \Delta K = 0$.

The tension $\vec{T}$ is perpendicular to the path: $W_T = 0$. Gravity: the child rises by $h = L(1 - \cos 35^\circ) = 3.30(0.181) = 0.597 \text{ m}$, so $W_w = -mgh$.

A child on a swing: solution

Set up: the child moves slowly, so K is almost constant: $W_{\text{tot}} = \Delta K = 0$.

The tension $\vec{T}$ is perpendicular to the path: $W_T = 0$. Gravity: the child rises by $h = L(1 - \cos 35^\circ) = 3.30(0.181) = 0.597 \text{ m}$, so $W_w = -mgh$.

Execute: $W_F + W_T + W_w = 0 \Rightarrow W_F = mgh = (25.0)(9.8)(0.597) = 1.5 \times 10^2 \text{ J}$.

A child on a swing: solution

Set up: the child moves slowly, so K is almost constant: $W_{\text{tot}} = \Delta K = 0$.

The tension $\vec{T}$ is perpendicular to the path: $W_T = 0$. Gravity: the child rises by $h = L(1 - \cos 35^\circ) = 3.30(0.181) = 0.597 \text{ m}$, so $W_w = -mgh$.

Execute: $W_F + W_T + W_w = 0 \Rightarrow W_F = mgh = (25.0)(9.8)(0.597) = 1.5 \times 10^2 \text{ J}$.

Evaluate: positive, as predicted (you push along the motion); it equals the work needed to raise the child through h . Using $F ds$ directly would need an integral over the changing force.

✓

Rule: power

$$P_{\text{av}} = \frac{\Delta W}{\Delta t} \quad P = \frac{dW}{dt} = \vec{F} \cdot \vec{v}$$

Power is the **rate** of doing work. Unit: watt, $1 \text{ W} = 1 \text{ J/s}$.

- $1 \text{ kW} = 10^3 \text{ W}$; $1 \text{ hp} = 746 \text{ W}$.
- The *kilowatt-hour* is a unit of energy:
 $1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$.

Common error

Power is not energy. A large energy can be delivered slowly (small power) or quickly (large power).

TOOLKIT 7 – Power

1. **AVERAGE POWER:** $P_{\text{av}} = W/\Delta t$ (total work over the time taken).
 2. **INSTANTANEOUS POWER:** $P = \vec{F} \cdot \vec{v} = Fv \cos \phi$ at one instant.
 3. **UNITS:** watt, kilowatt, horsepower; convert carefully.
 4. **COMPARE:** to find a time, use $t = W/P$; to find a force at constant P , use $F = P/v$.
- WATCH OUT:** power depends on how fast the work is done, not on the amount of work alone.
- CHECK:** is the power of a human of order 10^2 W , a car of order 10^4 W ?

Question

A 60 kg climber goes up a 4.0 m height of stairs in 5.0 s. (a) What is the average power developed? (b) An 80 kg climber develops the same power. How long does the same climb take?

Predict first

Will the heavier climber take longer or shorter?

Question

A 60 kg climber goes up a 4.0 m height of stairs in 5.0 s. (a) What is the average power developed? (b) An 80 kg climber develops the same power. How long does the same climb take?

Predict first

Will the heavier climber take longer or shorter?

(a) Work against gravity: $W = mgh = (60)(9.8)(4.0) = 2350 \text{ J}$.

$P = W/t = 2350/5.0 = 4.7 \times 10^2 \text{ W}$.

(b) $t = W/P = (80)(9.8)(4.0)/470 = 6.7 \text{ s}$.

Evaluate: the heavier climber is slower at the same power, as predicted. ✓

Question

A car moves at a constant 25.0 m/s on level ground against a total resistive force of 600 N . What power must the engine deliver to the wheels?

Predict first

Constant speed: what is the net force? How do the driving force and the resistive force compare?

Question

A car moves at a constant 25.0 m/s on level ground against a total resistive force of 600 N . What power must the engine deliver to the wheels?

Predict first

Constant speed: what is the net force? How do the driving force and the resistive force compare?

$\sum F = 0$, so the driving force equals the resistive force: $F = 600 \text{ N}$ along $\vec{v}$.
 $P = Fv = (600)(25.0) = 1.5 \times 10^4 \text{ W} = 15 \text{ kW} \approx 20 \text{ hp}$.

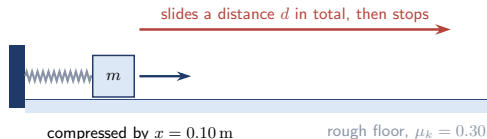
Evaluate: of order 10^4 W , as expected for a car. ✓

Integrated problem: a block and a spring on a rough floor

Decide what physics applies; do not look for a formula

A 0.50 kg block is pushed against a horizontal spring ($k = 800\text{ N/m}$), compressing it 0.10 m , and released from rest. The block is not attached. The floor is rough: $\mu_k = 0.30$ throughout.

- (a) How fast is the block moving as it leaves the spring (at natural length)?
- (b) How far does it slide in total before it stops?



Predict first

IDENTIFY $\rightarrow$ FBD and work of each force $\rightarrow$ PREDICT $\rightarrow$ EXECUTE $\rightarrow$ EVALUATE.

Identify, set up: the spring force varies (Toolkit 6); friction is constant (Toolkits 1 and 3); the work–energy theorem links the states. $f_k = \mu_k mg = 0.30(0.50)(9.8) = 1.47 \text{ N}$;
 $W_{\text{spring}} = \frac{1}{2}kx^2 = \frac{1}{2}(800)(0.10)^2 = 4.0 \text{ J}$.

Identify, set up: the spring force varies (Toolkit 6); friction is constant (Toolkits 1 and 3); the work–energy theorem links the states. $f_k = \mu_k mg = 0.30(0.50)(9.8) = 1.47 \text{ N}$;
 $W_{\text{spring}} = \frac{1}{2}kx^2 = \frac{1}{2}(800)(0.10)^2 = 4.0 \text{ J}$.

(a) From release to $x = 0$:

$$W_{\text{tot}} = 4.0 - (1.47)(0.10) = 3.85 \text{ J} = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2(3.85)/0.50} = 3.9 \text{ m/s}.$$

Identify, set up: the spring force varies (Toolkit 6); friction is constant (Toolkits 1 and 3); the work–energy theorem links the states. $f_k = \mu_k mg = 0.30(0.50)(9.8) = 1.47 \text{ N}$;

$$W_{\text{spring}} = \frac{1}{2}kx^2 = \frac{1}{2}(800)(0.10)^2 = 4.0 \text{ J}.$$

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$$W_{\text{tot}} = 4.0 - (1.47)(0.10) = 3.85 \text{ J} = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2(3.85)/0.50} = 3.9 \text{ m/s}.$$

(b) From release to rest ($K_2 = 0$): $W_{\text{tot}} = 4.0 - f_k d = 0 \Rightarrow d = \frac{4.0}{1.47} = 2.7 \text{ m}$ in total.

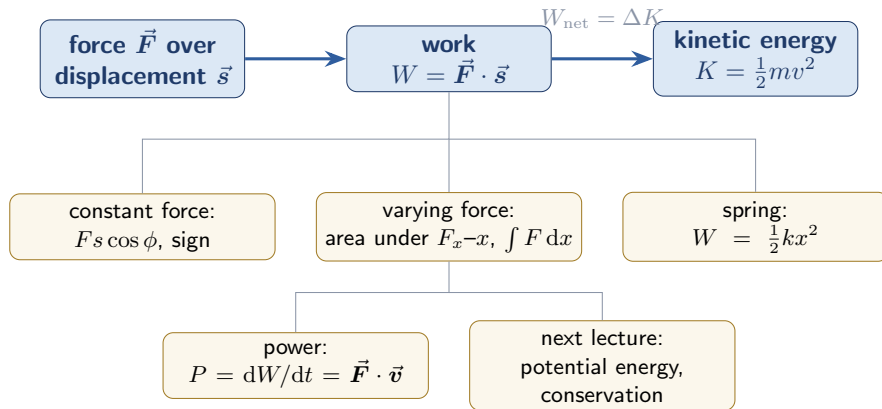
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(a) From release to $x = 0$:

$$W_{\text{tot}} = 4.0 - (1.47)(0.10) = 3.85 \text{ J} = \frac{1}{2}mv^2 \Rightarrow v = \sqrt{2(3.85)/0.50} = 3.9 \text{ m/s}.$$

(b) From release to rest ($K_2 = 0$): $W_{\text{tot}} = 4.0 - f_k d = 0 \Rightarrow d = \frac{4.0}{1.47} = 2.7 \text{ m}$ in total.

Evaluate: after leaving the spring the block slides $2.7 - 0.10 = 2.6 \text{ m}$, and friction does $-(1.47)(2.6) = -3.85 \text{ J}$, exactly removing the kinetic energy from (a). ✓



MASTER PROCEDURE – ENERGY

1. **IDENTIFY** The object, and what is asked: a speed, a distance, a force, a work or a power.
 2. **STATES** Mark the initial and final positions with their speeds.
 3. **DRAW THE FBD** All forces on the object, and the displacement (Week 3).
 4. **WORK OF EACH FORCE** Sign first, then size: constant, graph, integral or spring (Toolkits 1 to 6).
 5. **PREDICT** Sign of W_{tot} , speeding up or slowing down, rough size.
 6. **SELECT THE METHOD** Energy method or Newton's second law (Toolkit 4).
 7. **EXECUTE** $W_{\text{tot}} = \Delta K$; add power if a rate is asked (Toolkit 7).
 8. **EVALUATE** Units, sign of ΔK , check by a second method.
- SKETCH $\rightarrow$ PREDICT $\rightarrow$ CALCULATE $\rightarrow$ CHECK.

Inside the course framework

IDENTIFY $\leftrightarrow$ steps 1–2
SET UP $\leftrightarrow$ steps 3–6
EXECUTE $\leftrightarrow$ step 7
EVALUATE $\leftrightarrow$ step 8

Physical meaning

Next lecture: potential energy turns the work done by gravity and springs into stored energy, and leads to conservation of energy.



THE SEVEN TOOLKITS

- | | | |
|---|---------------------------------------|---|
| 1 | Work by a constant force | force and displacement; angle ϕ ; sign; $W = F s \cos \phi$ |
| 2 | Net work from the FBD | list forces; sign each; Method I and Method II |
| 3 | The work–energy theorem | two states; forces doing work; $W_{\text{tot}} = \Delta K$ |
| 4 | Choosing the method | energy for speed and distance; Newton for acceleration and time |
| 5 | Work from graphs and integrals | area under F_x – x ; $\int F_x dx$ |
| 6 | Springs | Hooke's law; $W = \frac{1}{2} k x^2$; signs by and on the spring |
| 7 | Power | $W/\Delta t$ and $\vec{F} \cdot \vec{v}$; watt, kW, hp |

The Master Procedure sits above these. Week 1 to 3 toolkits (dot product, components, free-body diagrams, friction) are reused, not renamed.

Work (constant force)

$$W = \vec{F} \cdot \vec{s} = F s \cos \phi$$

Total work

$$W_{\text{tot}} = W_1 + W_2 + \cdots = (\sum \vec{F}) \cdot \vec{s}$$

Work–energy theorem

$$W_{\text{tot}} = \Delta K, \quad K = \frac{1}{2} m v^2$$

Work by the weight

$$W_w = \pm mgh \text{ (down +, up -)}$$

Varying force

$$W = \int_{x_1}^{x_2} F_x dx \text{ (area)}$$

Spring

$$F = kx, \quad W = \frac{1}{2} k x_2^2 - \frac{1}{2} k x_1^2 \text{ (on the spring)}$$

Power

$$P_{\text{av}} = W/\Delta t, \quad P = \vec{F} \cdot \vec{v}$$

Six habits for Week 4

✓ Habit 1

Draw the free-body diagram and the displacement first.

✓ Habit 2

Decide the sign of each work before calculating it.

✓ Habit 3

Use the net work with the work–energy theorem, not one force’s work.

✓ Habit 4

No displacement, or a perpendicular force, means no work.

✓ Habit 5

Measure the spring stretch from its natural length.

✓ Habit 6

Check by a second method, and compare with your prediction.

Checklist: can I ...

- ☐ calculate work as $\vec{F} \cdot \vec{s} = F s \cos \phi$?
- ☐ decide the sign of work from the angle?
- ☐ list the forces on a free-body diagram that do work?
- ☐ find total work by two methods?
- ☐ state the work–energy theorem and use $K = \frac{1}{2}mv^2$?
- ☐ explain why $K \propto v^2$?
- ☐ choose between the energy method and Newton's second law?
- ☐ find work from an F_x – x graph or an integral?
- ☐ calculate the work done by or on a spring?
- ☐ explain why a perpendicular force does no work?
- ☐ calculate average and instantaneous power?
- ☐ select the appropriate model in an unfamiliar problem?

After this week

Tutorial sheet: extra work, work–energy, spring, friction and power problems.
Supplementary notes: the dot-product refresher, the proof of the theorem for variable forces, the work along a general path, and what happens to the energy lost to friction.